



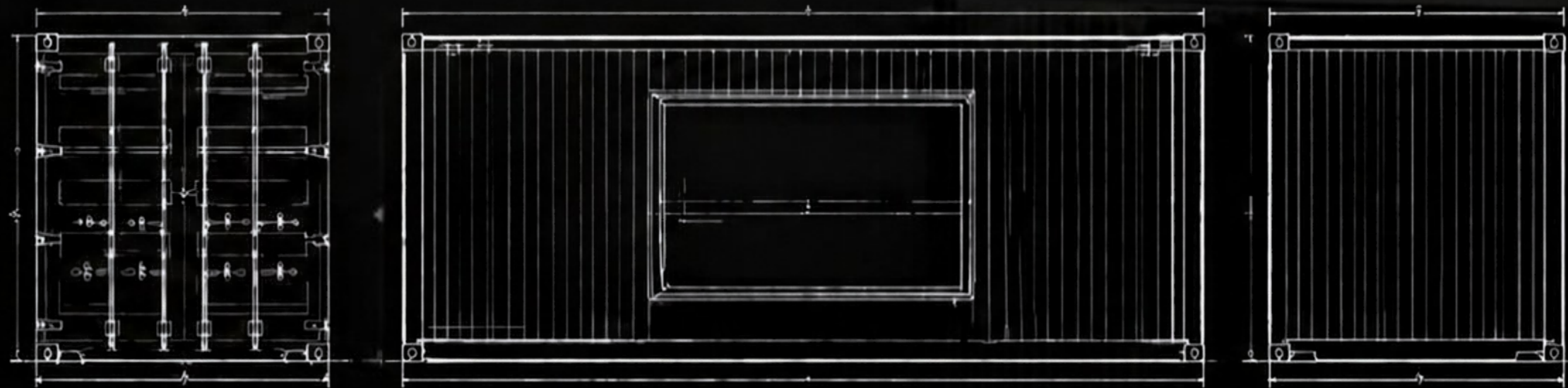
PROJECT STORYBOARD // CASE 5

SSU CONNEX

CONTAINER CONVERSION

CUSTOM METAL FABRICATION PROJECT

Fayetteville, North Carolina • Delivered 2



VETERAN OWNED • AMERICAN

CLIENT SECTOR

Logistics / Mobile Ops

SCOPE

Structural Conversion

DELIVERY

Q2 - 2026

BUILT STRONG. BUILT RIGHT.

A shipping container, reengineered from the frame out.

OBJECTIVE

Convert a standard SSU Connex container into a mission-ready structural enclosure with reinforced floor walls, and overhead framing.

CUSTOMER NEED

A durable, transportable steel structure engineered for repeat deployment and heavy interior loading • beyond factory tolerances.

METAL WROX SOLUTION

In-house fabrication of a fully welded floor system, full length wall supports, overhead reinforcement, and interior steel lining assembled, test fit, and installed by our veteran crew.

PROJECT STATUS

Delivered • On schedule • Client accepted



PROJECT FACTS

CONTAINER

20-ft SSU Connex

SCOPE

Floor • Walls • Overhead • Liner

MATERIAL

A36 structural steel

CLIENT SECTOR

Logistics / Mobile Ops

MAJOR MILESTONES



PHASE 01 - 10.14.2026

CONNEX ARRIVAL AND INITIAL INSPECTION

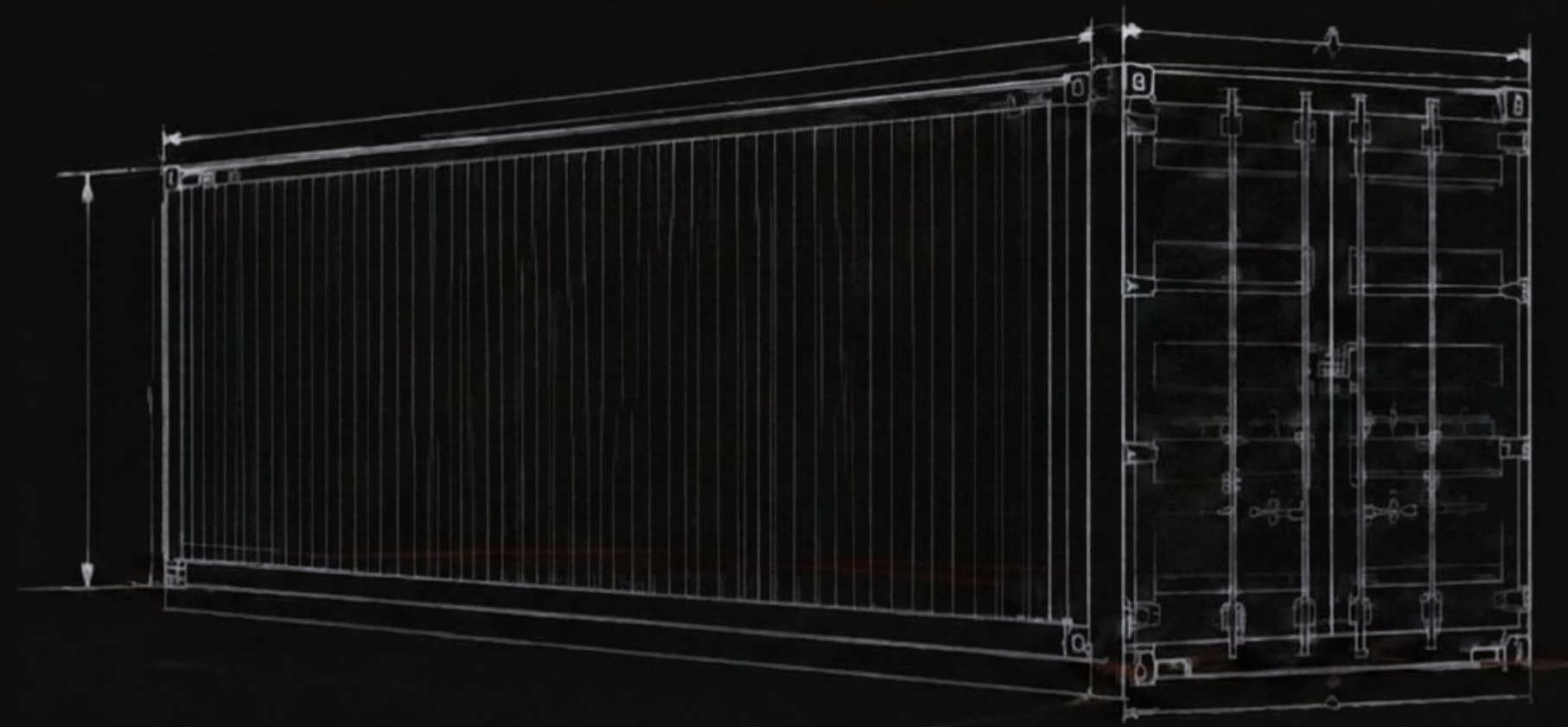
The SSCU Connex arrived at our Lafayetteville shop in factory-standard configuration — corrugated steel skin, plywood decking, minimal internal framing. Our inspection documented weld seams, deck condition, and dimensional tolerances against the customer's build package.

Field notes flagged the standard floor as under-rated for the intended payload, and the sidewall ribs as insufficient anchor points. The transformation plan calls for a purpose-built structural system fabricated from A36 steel, welded in place and finished to marine-grade standards.

CAPTION Container as received — original condition documented in a full 360° walkaround.

Deliverables engineered to structural intent.

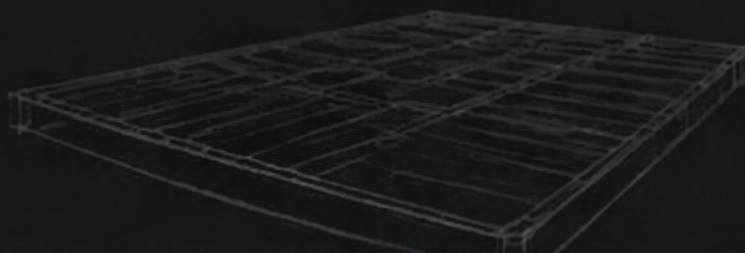
Four fabrication packages define the conversion. Each is drawn, cut, and welded in-house before all.



01

Structural Floor

Full-length steel floor system fabricated from A36 plate and channel, seam-welded to the container base rails.



02

Wall Supports

Vertical wall supports tied into the corner posts, tensioned and gusseted for point-load resistance.



03

Overhead Reinforcement

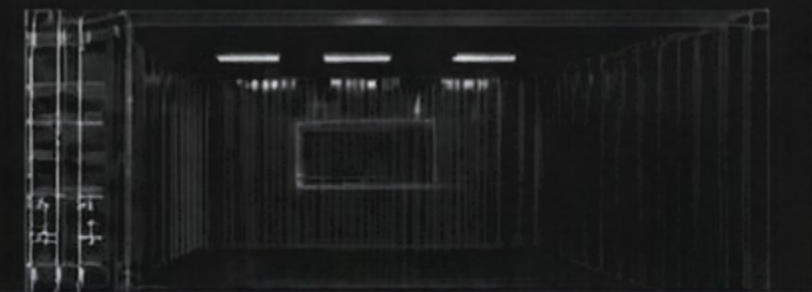
Overhead cross-members and load headers to eliminate roof deflection under interior rigging.



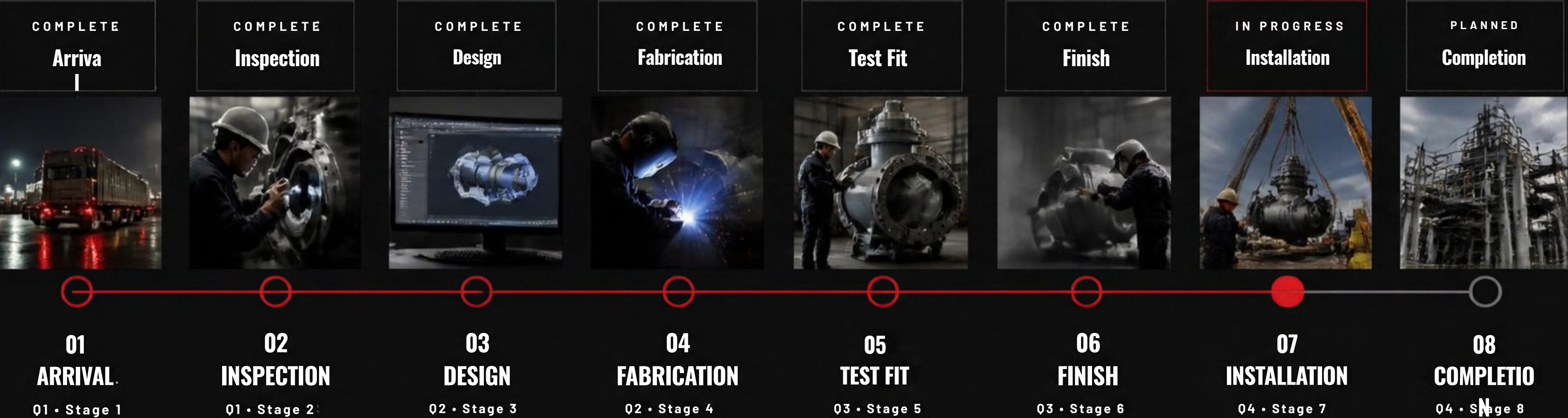
04

Interior Steel Lining

Full-perimeter lining plate, welded flush and prepped for coating.



From arrival to installation.



CURRENT PHASE • COMPLETED •



PHASE 03

03

WALL SUPPORTS & OVERHEAD REINFORCEMENT

Full-length wall supports are fabricated and tied into the corner posts, while overhead reinforcement locks the top plane against roof deflection.



PROGRESS

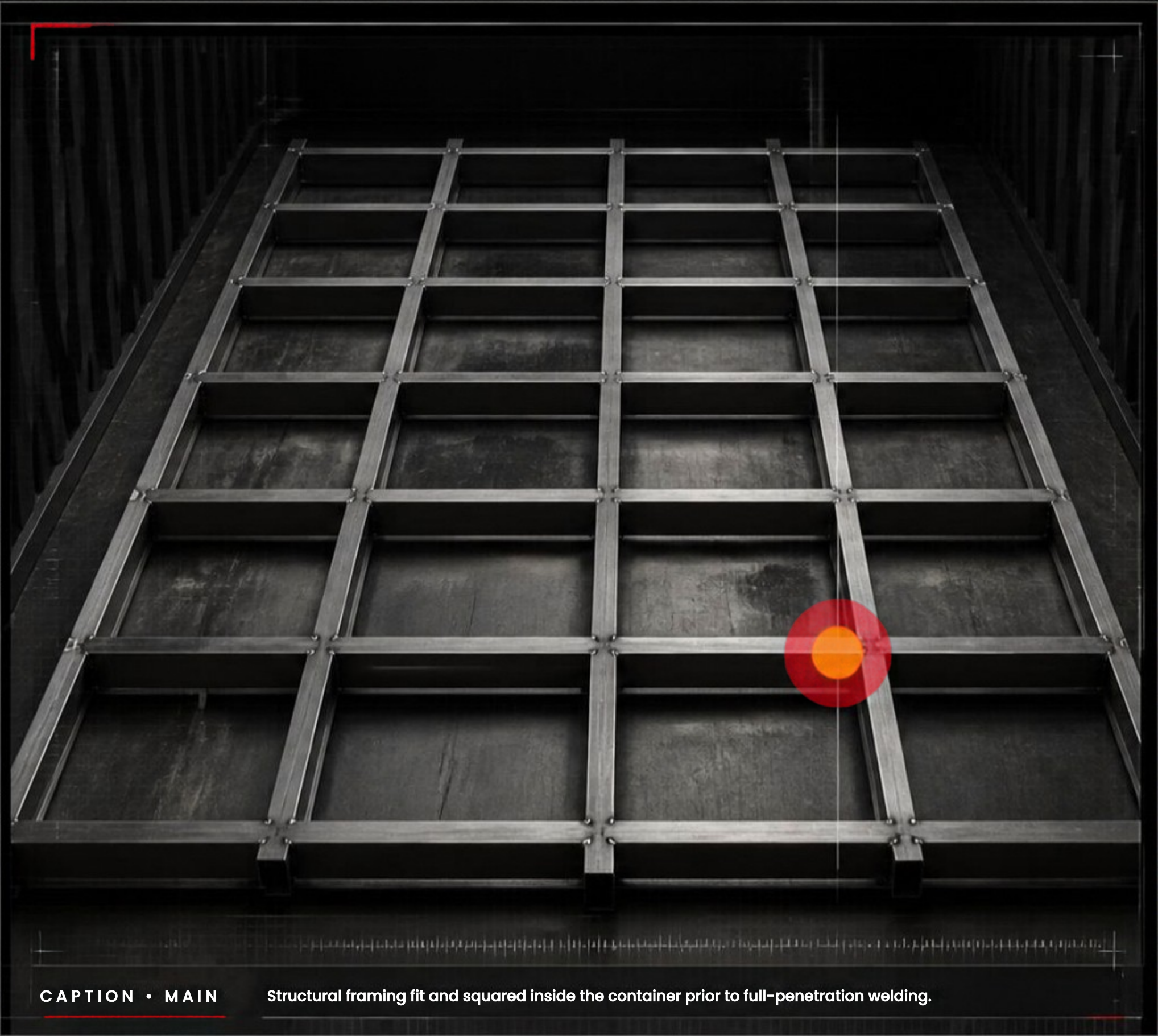
50% • STAGE 4 OF 8

Documenting the Original Interior



CAPTION

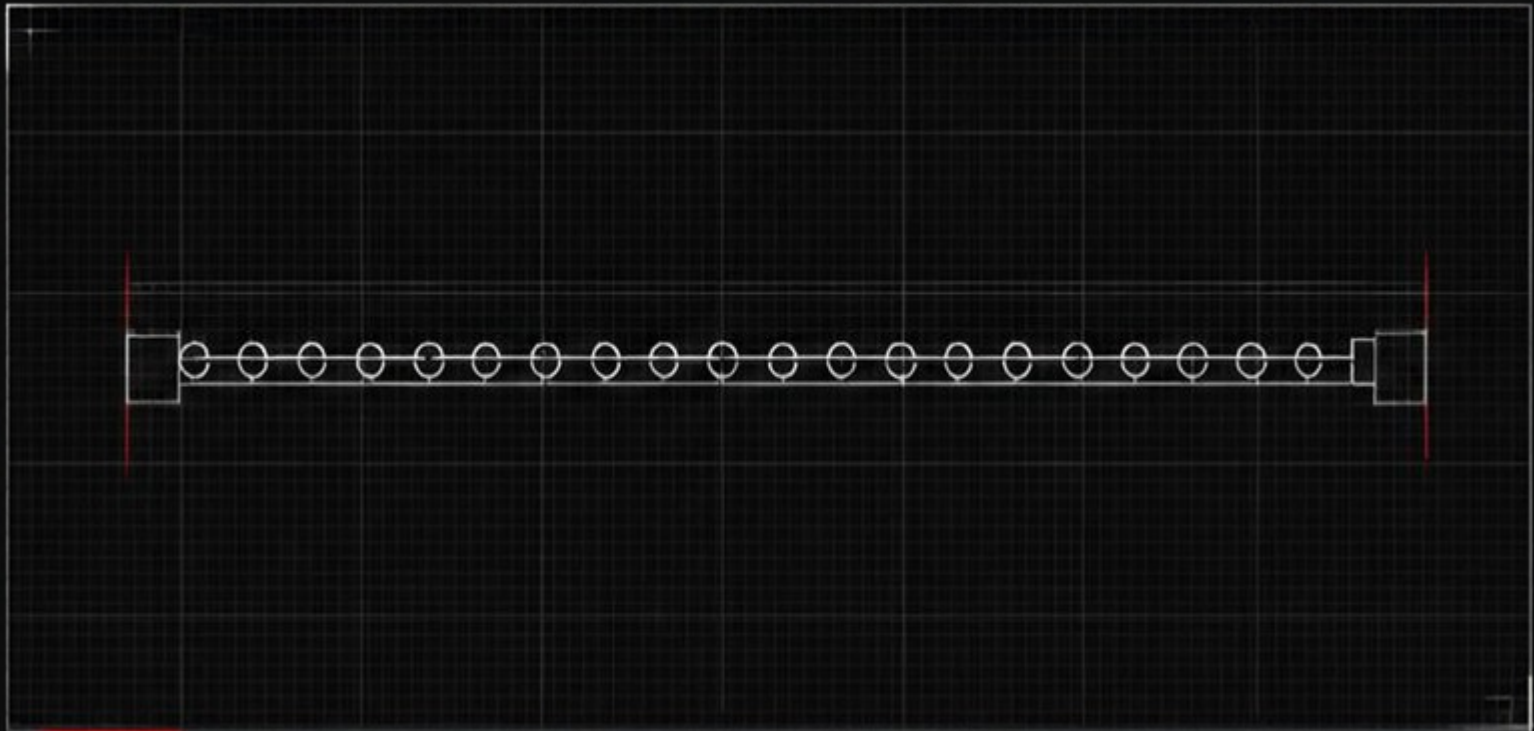
Interior baseline photo prior to fabrication — plywood decking, corrugated skin, factory ribs.



CAPTION • MAIN

Structural framing fit and squared inside the container prior to full-penetration welding.

PHASE 02 • STRUCTURAL FOUNDATION



DETAIL MIG weld • root pass, A38 plate to base rail.

Building the structural foundation.

The floor system starts here. A38 channel and plate are cut, squared, and tack-fit inside the container before final passes (back the frame to the base rail).

Steel: A38 channel + plate
Prep: Squared to .11")16" •

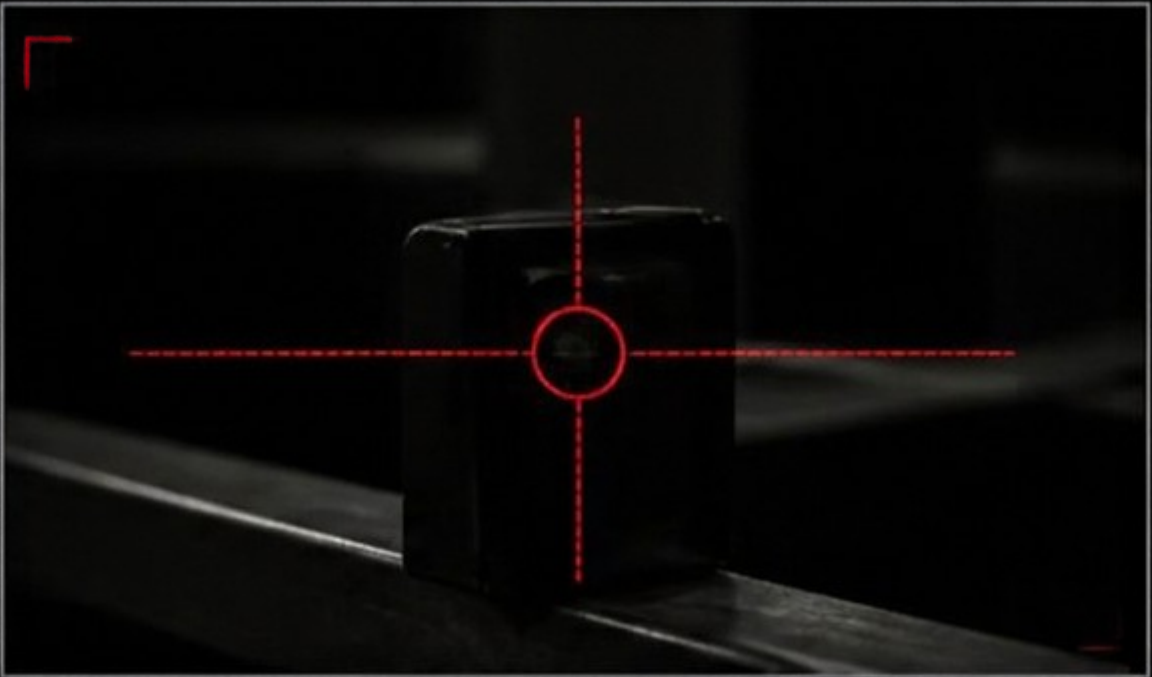
Process: MIG (GMAW), C45 ER7D5-G
Paste: Foot + fill + cap • inspection Visual • dye
pericrane

Steel Floor System — Installed & Squared.



POSITIONING - SQUARE - WED

01 • Positioning the fabricated floor system.
The floor system anchors every subsequent build stage. With the frame squared and locked, wall supports and overhead reinforcement can be indexed off a true datum • cutting install variance from the outlet.



02 • Alignment check, x1/16 •



03 • Weld pass on floor rail.

PHASE 03

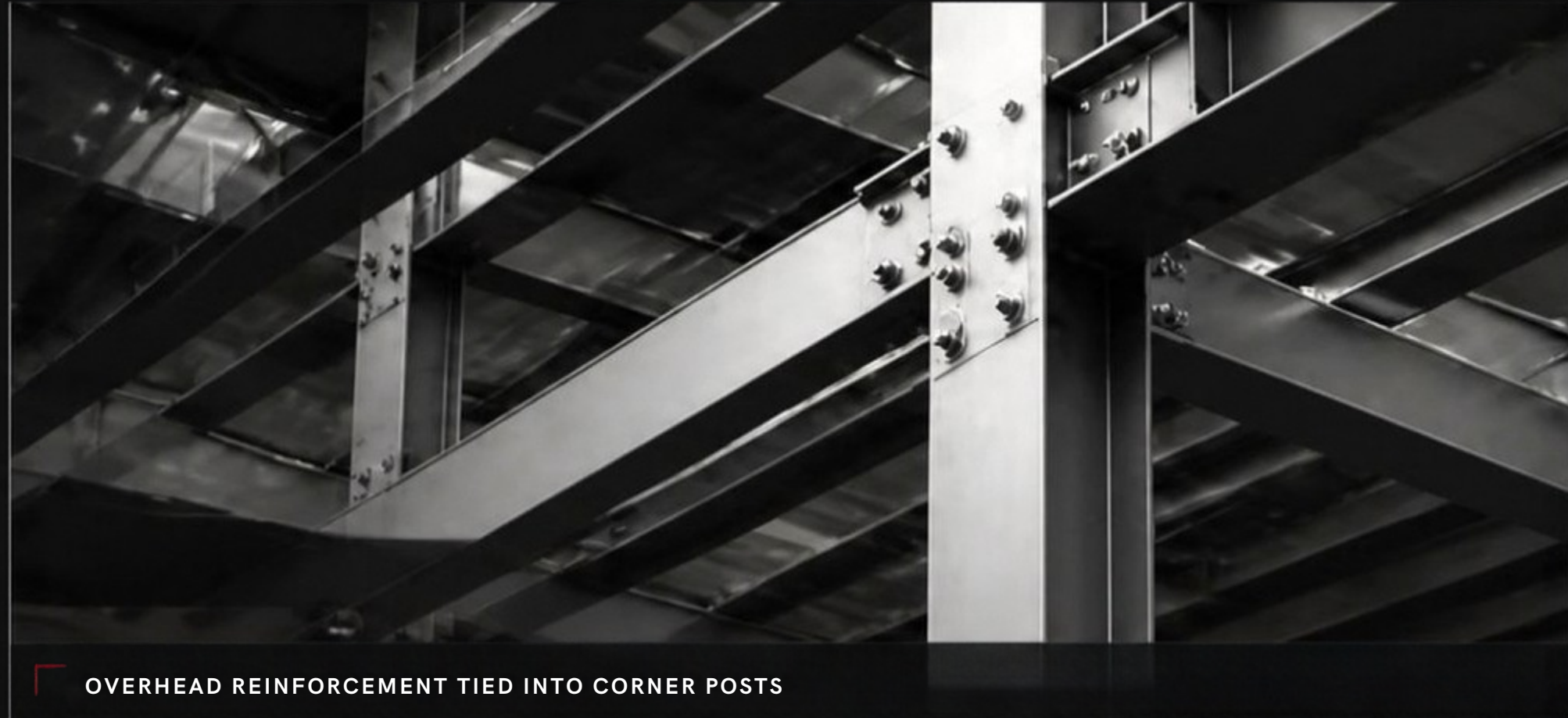
03

Wall Supports & Overhead Reinforcement

Full-length wall supports are fabricated and tied into the corner posts, while overhead reinforcement locks the top plane against roof deflection.

PROGRESS

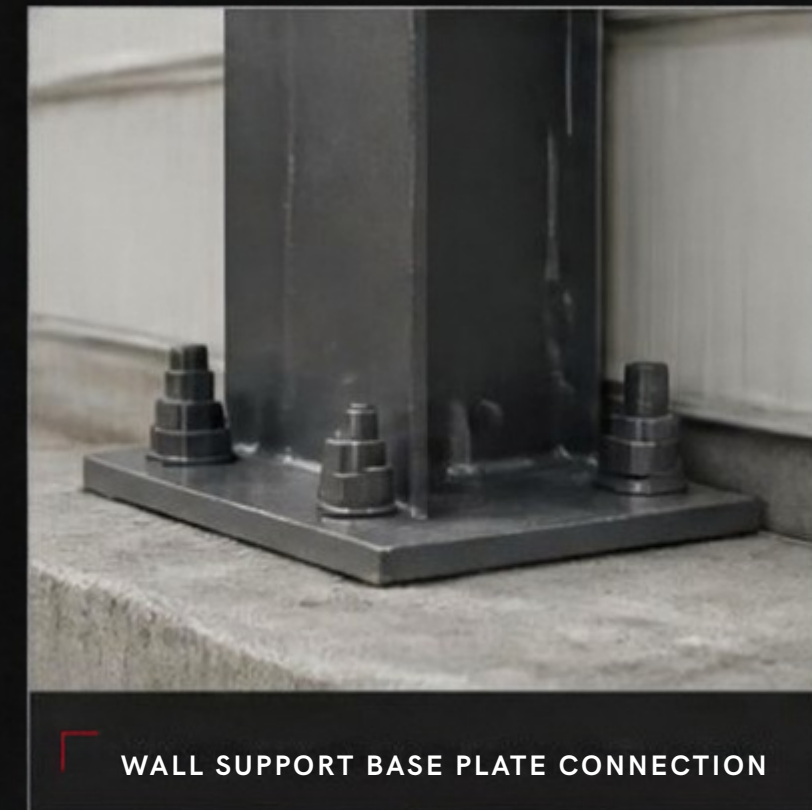
50% • Stage 4 of 8



OVERHEAD REINFORCEMENT TIED INTO CORNER POSTS

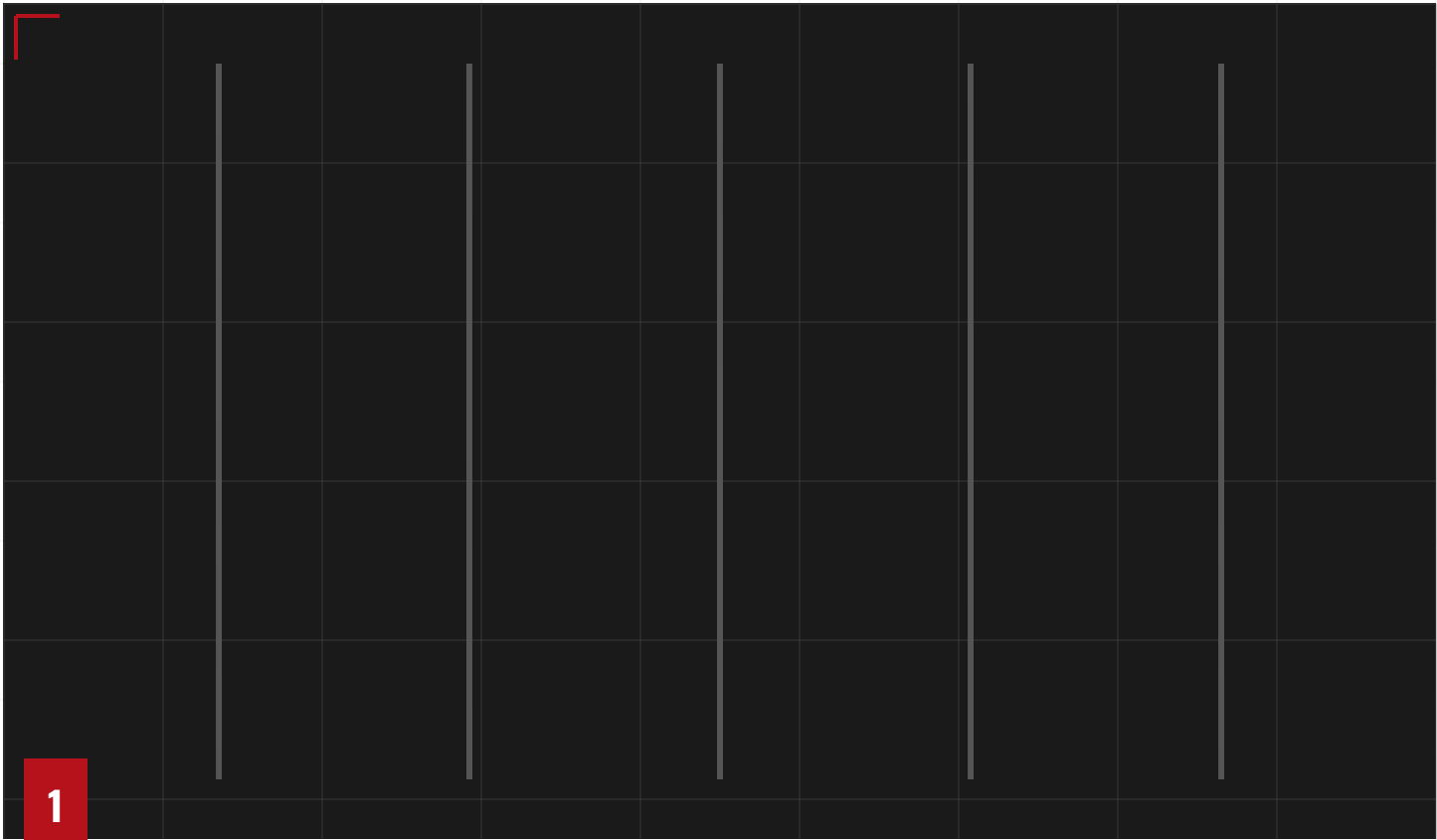


FULL-LENGTH WALL SUPPORTS AT REGULAR INTERVALS

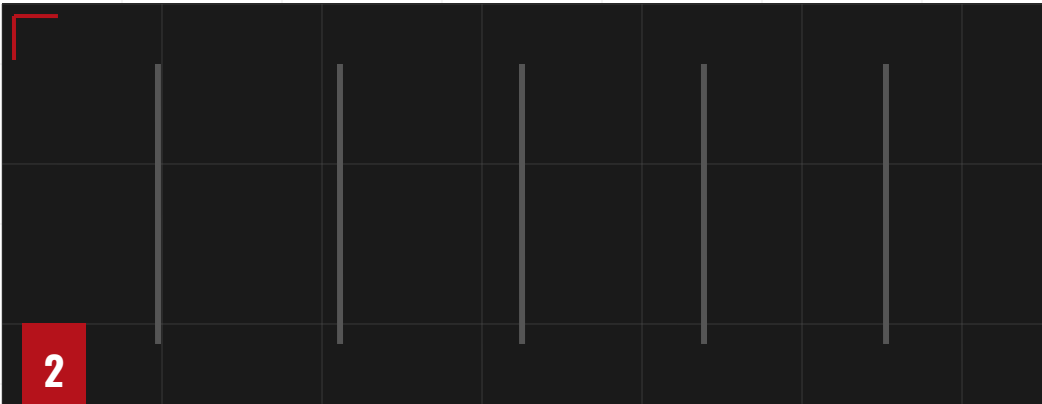


WALL SUPPORT BASE PLATE CONNECTION

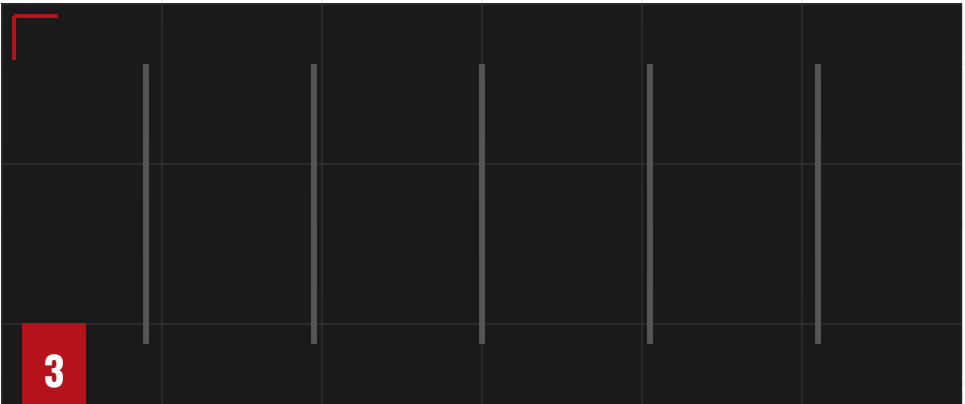
Installing Full-Length Wall Supports.



01 Full-length support tacked plumb along the sidewall — corner posts referenced.



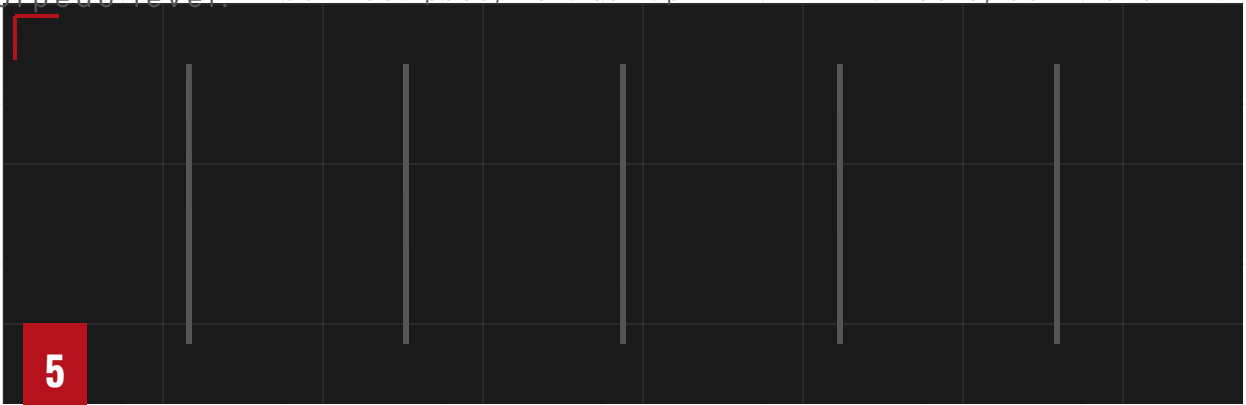
02 Alignment check with laser plumb and torpedo level.



03 Root pass, vertical up — E71T-1 flux core, downhand fill.



04 Crew hand-off between weld positions.



05 Support run complete, ready for gusset plates.

FIELD CHALLENGES & SOLUTIONS

CHALLENGE

Corner posts out of plumb by ~3/8" over full height, biasing every support run.

SOLUTION

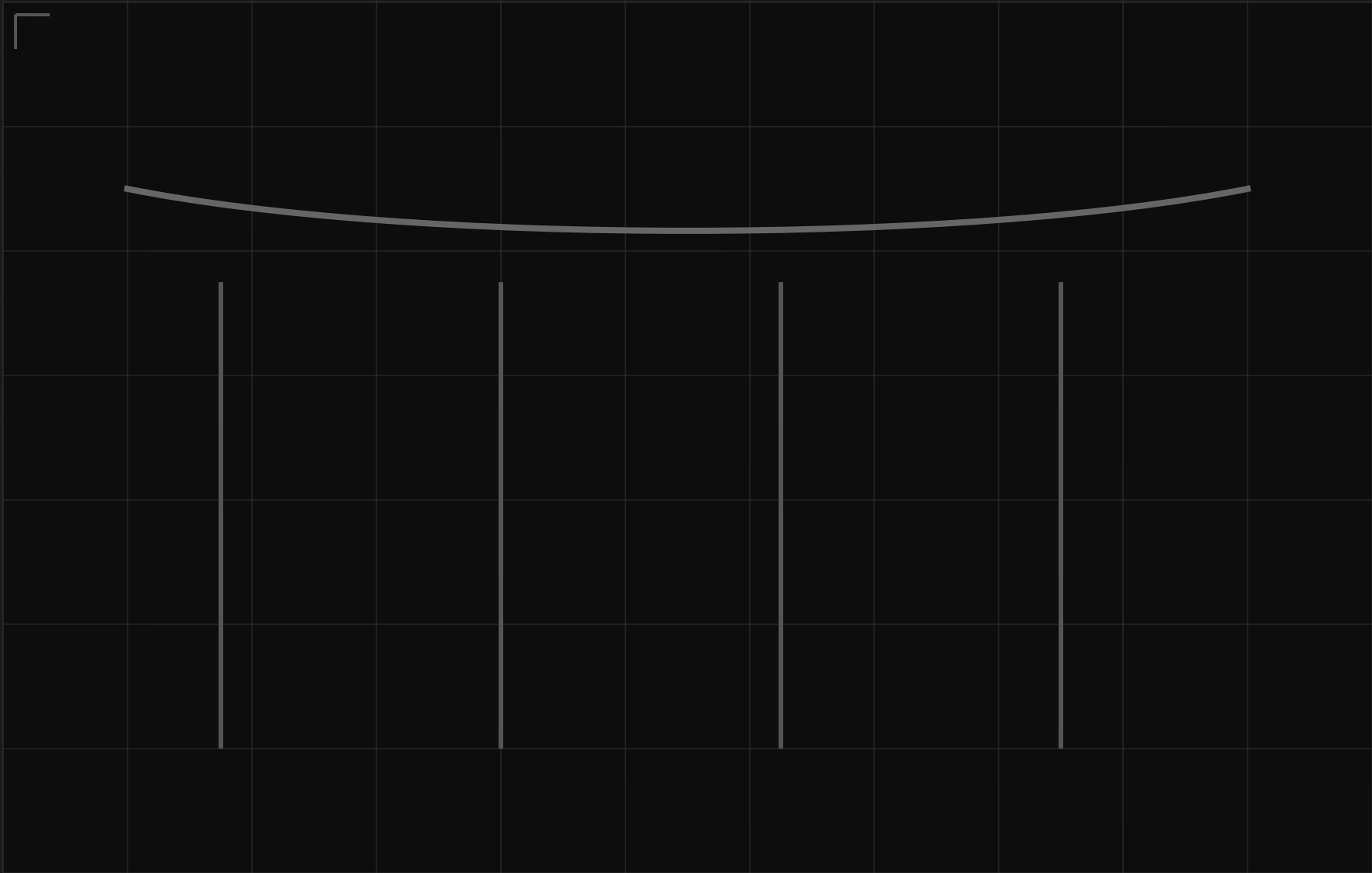
Fabricated adjustable gusset plates to shim each support at the top and bottom rail.

SAFETY

Hot-work permit, forced-air ventilation, and dual fire watch throughout all vertical positions.

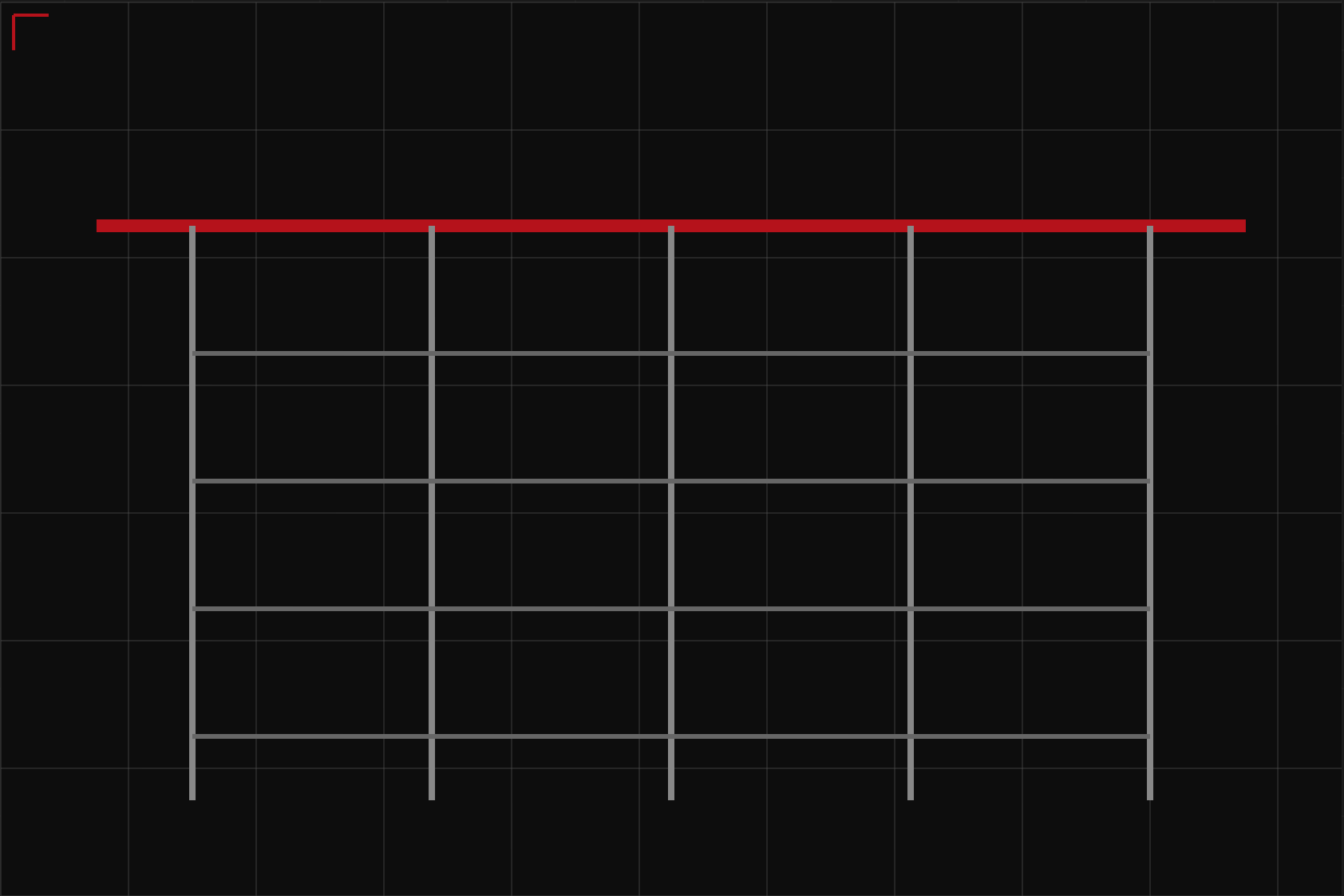
Reinforcing the Upper Structure.

BEFORE



Deflection observed under load; roof plane unrestrained between corner posts.

AFTER



Cross-header and paired stiffeners tied into wall supports — plane locked, deflection eliminated.

TRANSFORMATION SUMMARY

Overhead deflection: 3/8" → 0. Load rating: doubled. Interior rigging: sanctioned for full spec.

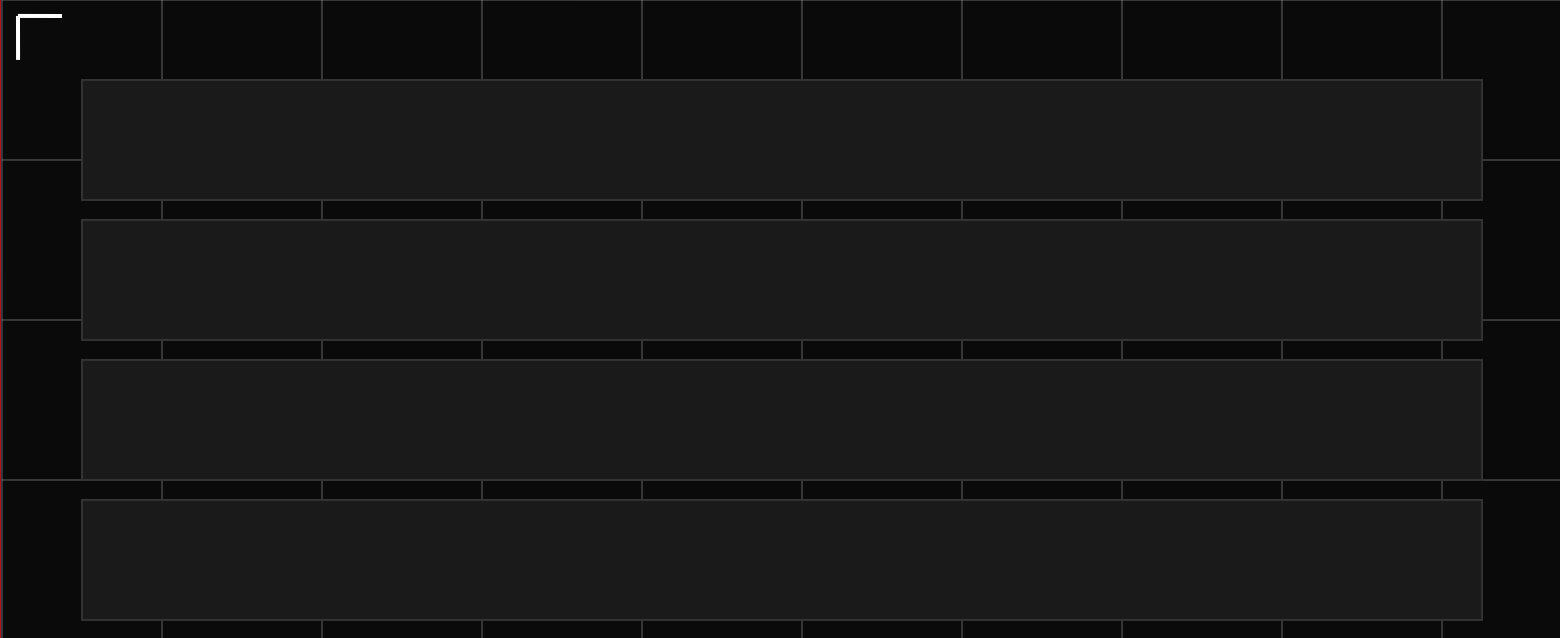
Skinning the interior in welded plate.



CHALLENGE

Uneven substrate.

Original ribbed skin creates a discontinuous mounting surface — full-perimeter lining plate must span factory undulations without oil-canning or losing weld access to the corner posts.



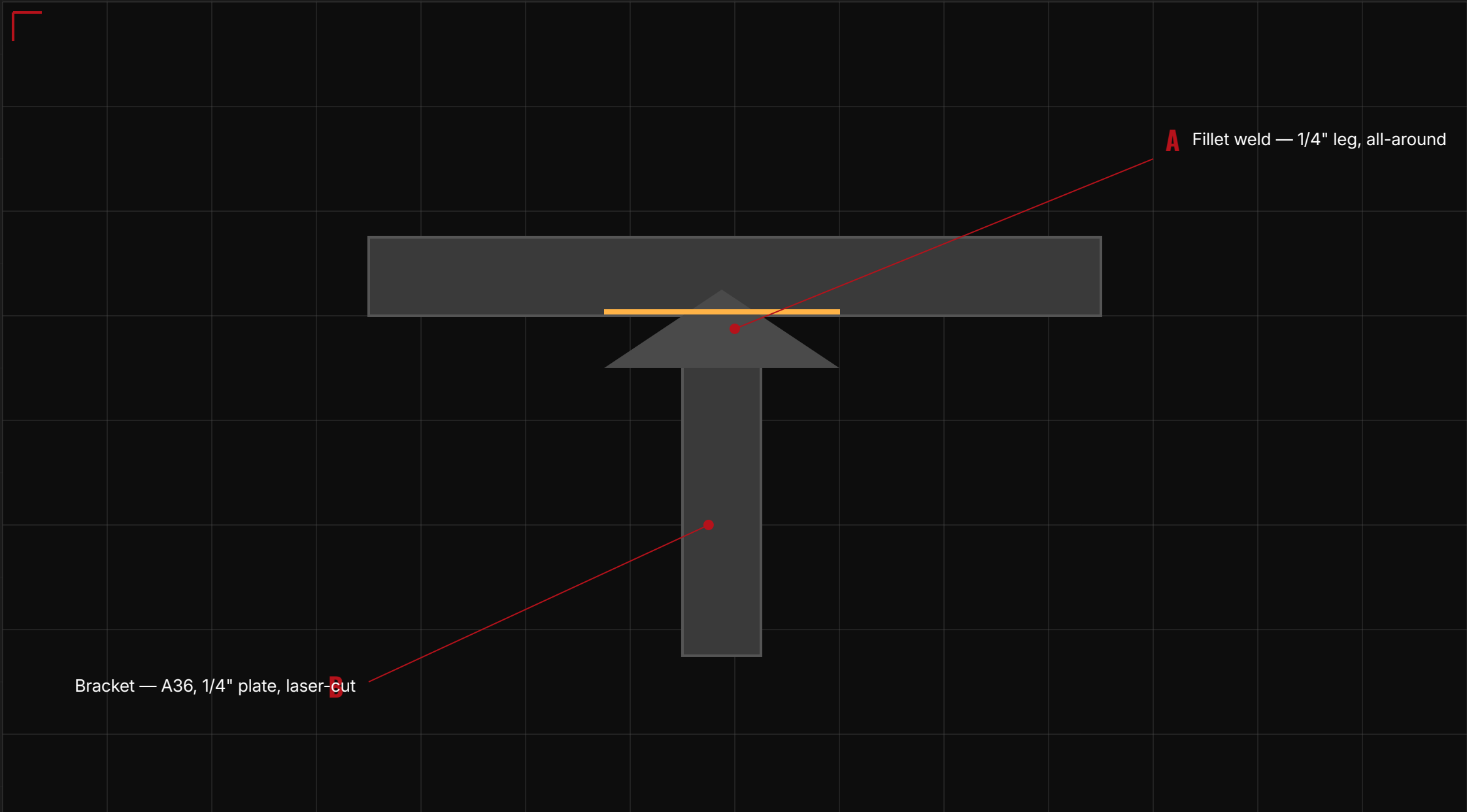
METAL WORX SOLUTION

Furring, plate, seam-weld.

Fabricated furring strips index off the wall supports to build a true plane. 14-ga plate is fitted, stitch-welded, then continuous-seamed and ground flush — a monolithic interior ready for coating.

CRAFTSMANSHIP CALLOUT · Continuous seam · zero oil-can · coating-ready

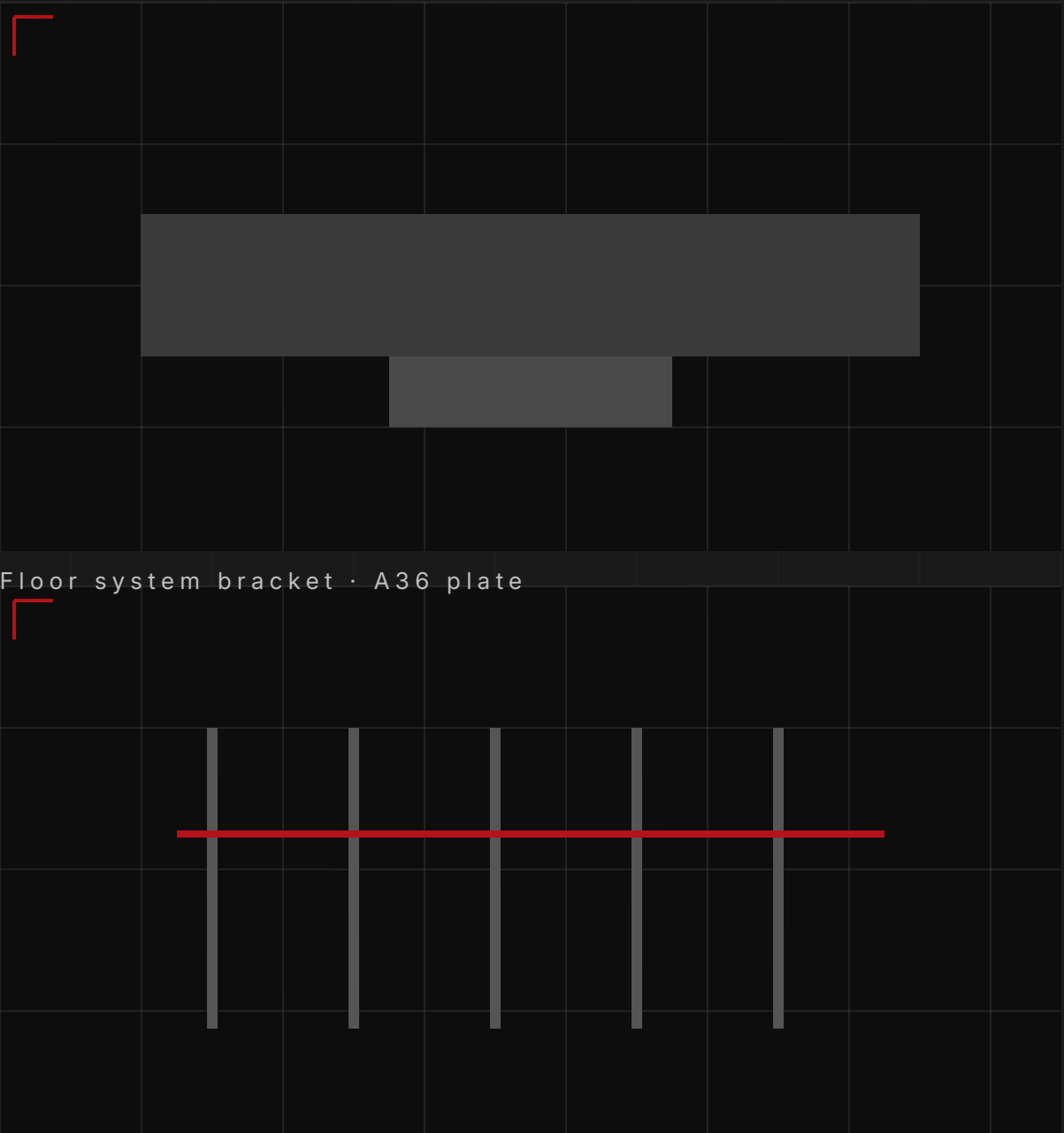
Where the welds do the talking.



A Fillet weld — 1/4" leg, all-around

Bracket — A36, 1/4" plate, laser-cut

Welded bracket assembly — floor rail to vertical stiffener, gusseted, 1/4" fillet all-around.



Floor system bracket · A36 plate

Wall reinforcement · paired stiffener + header

Every stage - checked and closed.

01

Arrival and inspection

02

Structural foundation fabricated

03

Wall supports installed

04

Overhead reinforcement finished

05

Interior lining completed

06

Test fitting and alignment

07

Installation finalized

08

Project completed

Boots on the ground. On-site, on-spec.



CREW

5-person install team, all trades on-site.

EQUIPMENT

Rough-terrain forklift · MIG rig · plasma cutter.

SAFETY

Fire watch · hot-work permit · PPE tier 3.

SITE CONDITIONS

Concrete pad, 42° F, mild wind — heat blankets deployed.

CHALLENGE OVERCOME

Field-verified corner variance; on-site shims fabricated in ≈ 40 minutes.

PROJECT COMPLETE · Q2 · 2026

SSU Connex — Delivered.

Custom Metal Fabrication · Fayetteville, NC



Built to endure — the finished conversion passed client acceptance on the first walkthrough. Reinforced, sealed, and coating-ready.

RESULTS · DELIVERED

Custom Metal. Built to Last.

DELIVERY

On time · On spec · Zero rework

CRAFTSMANSHIP

Full-penetration welds · square to $\pm 1/16''$

IMPACT

Load rating doubled · client re-order authorized

VETERAN OWNED

AMERICAN MADE

BUILT STRONG. FINISHED RIGHT.

